

PUSHKAR KODIGEHALLI UMESH

Nottingham, UK | Right to work in the UK | 07818147013 | pushkar.uk@outlook.com
linkedin.com/in/pushkarku | github.com/pushkarx17 | pushkarku.com

SUMMARY

Postgraduate in Electronic Communication and Computer Engineering (MSc, University of Nottingham; dissertation submitted, result pending) with applied machine learning, data and software experience. Built Python data pipelines and ML models on real GNSS and sensor data as a GeoAI intern, benchmarked regression and classification models for an explainable-AI dissertation, implemented perceptron learning from scratch in MATLAB, and developed a SwiftUI/SwiftData iOS app. Completed GeoAI and systems operations internships in 2026.

SKILLS

- **Languages:** Python, MATLAB, Swift, SQL (MySQL), C, C++, HTML, CSS
- **Machine Learning:** Supervised Learning, Regression, Classification, Random Forest, Gradient Boosting, SVM, Logistic Regression, Model Evaluation, Explainable AI, Neural Networks (perceptron)
- **Data:** Data Pipelines, Data Cleaning, Data Analysis, Statistical Analysis, NumPy, Pandas, Matplotlib, QGIS, SNAP
- **Mobile & Tools:** SwiftUI, SwiftData, iOS Development, Xcode, Git, GitHub, AWS (fundamentals), Linux (basic)

PROJECTS

Explainable AI for GNSS Data Quality and Environment Assessment from NMEA Logs

MSc Dissertation (submitted), University of Nottingham, supervised by Dr Minglei You

2026

- Benchmarked regression (Ridge, SVM, Random Forest, Gradient Boosting) and classification (Random Forest, Gradient Boosting, SVM, Logistic Regression) models on GNSS NMEA logs.
- Compared the models against HDOP-only (both tasks), training-mean (regression) and majority-class (classification) baselines; applied explainable AI to assess data quality and environment.

LivingSolo iOS App, Personal Project

Jan 2026 – present

- Built an iOS app for people living alone, covering kitchen inventory, expenses and to-do lists, using SwiftUI TabView navigation and SwiftData for persistent local storage, and following iOS Human Interface Guidelines.

Perceptron Neural Network from First Principles, University of Nottingham – Coursework

2026

- Implemented the Perceptron Learning Law from scratch in MATLAB, with manual weight updates and convergence verified on a binary NOR gate classifier, then extended it to 4-class classification using a one-vs-rest strategy.
- Analysed decision boundaries and misclassification rates, plotting model performance across training iterations.

WORK EXPERIENCE

GeoAI Intern, Geospatial Ventures Limited (GVL), Nottingham

Jun – Aug 2026

- Built Python pipelines parsing RINEX, NMEA and sensor data for a soil moisture and battery monitoring deployment.
- Diagnosed a base station failure via backtest analysis; ran statistical analysis and outlier removal on 22 days of fix-quality data.
- Produced geospatial visualisations in QGIS, SNAP and Matplotlib; developed and evaluated ML models for soil moisture and station uptime prediction; automated Sentinel-2 download and spectral index processing.

Systems Operationalise & SEO Campaign Manager (Internship), ICN Media, Nottingham

Aug – Sep 2026

- Authored Standard Operating Procedures, optimised workflows and audited website SEO performance.

Data Engineer (Micro Placement), University of Nottingham x Practera

Jun 2026

- Audited and reconciled two spreadsheet-based inventory systems, profiling 300+ records across 40+ fields to quantify data quality, field adoption and schema drift.
- Developed a framework classifying 40+ fields as Keep, Remove, Reintroduce or Modify to drive consolidation into a simplified data model.

EDUCATION

MSc Electronic Communication and Computer Engineering, University of Nottingham 2025 – 2026 (result pending)

Modules: Artificial Intelligence and Intelligent Systems (**77%**); Advanced Computational Engineering (**85%**); Scalable Cross-Platform Software Design (**76%**); HDL for Programmable Devices (**83%**).

BSc Physics, Mathematics and Electronics, CHRIST (Deemed to be University), India

2021 – 2024

CGPA 8.49/10 (First Class). Grades of A+ in Linear Algebra; Embedded Systems and IoT Fundamentals (C programming); Digital Signals and System Architecture; Data Communication and Networking.